MILITARY SPECIFICATION
CARTRIDGE, CALIBER .50, PRACTICE, T249E2

1. SCOPE

1.1 This specification covers cartridges intended for function tests of Caliber .50 Spotter-Tracer Weapons.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inpsection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-636	- Visual Inspection Standards for Small Arms Ammunition through Caliber .50
MIL-STD-644	- Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition

DRAWINGS

Munitions Command

D10534667	- Cartridge, Caliber .50, Practice, T249E2
C7643674	- Classification of Cartridge Case Defects
F10534780	- Packing and Marking, Cartridges, Caliber .50, Practice, T249E2
IEL-10534667	- Inspection Equipment List for Cartridges, Caliber .50, Practice, T249E2

PUBLICATIONS

Munitions Command

AMCR 700-700, Vol. III	- Manual of Test Methods for Small Arms Ammunition

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 General.- The cartridge shall comply with the requirements specified on Drawing D10534667, referenced specifications and the following:

3.2 Initial production sample.- Unless otherwise directed by the contracting officer, an initial production sample shall be required.

3.3 Bullet extraction.- The force required to extract the bullet from the cartridge case shall be not less than 200 pounds.

3.4 Residual stress.- The cartridge case shall not split when subjected to a 1.0% mercurous nitrate solution for 15 minutes.

3.5 Waterproof.- The cartridge shall not release more than one bubble of air when subjected to a pressure differential of 7-1/2 pounds per square inch (psi) for 30 seconds.

3.6 Velocity.- The average velocity of the sample cartridges, conditioned at 68 to 72° Fahrenheit (F), shall be 1745 feet per second (ft/sec) plus or minus 20 ft/sec at 78 feet from the muzzle. The standard deviation of the velocities shall not exceed 14 ft/sec.

3.7 Chamber pressure.- The average chamber pressure of the sample cartridges, conditioned at 68 to 72° F, shall not exceed 38,000 pounds per square inch (psi).

3.8 Accuracy.- The average of the mean radii of all targets of the sample cartridges fired at 600 yards shall not exceed 5 inches.

3.9 Function and casualty.- The cartridge shall function without casualty.

3.10 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.10.1 Metal defects.- The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents, and other defects.

3.10.2 Foreign matter.- The cartridge shall be free of corrosion, stains, discolorations, dirt, oil, and smears of lacquer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right

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to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.2 Initial production sample.- Prior to the initiation of regular production, the supplier shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of 500 cartridges. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as will be used in regular production. All parts and materials, including packaging and packing, must be the same as will be used in regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Lot identification.- Each lot of ammunition shall be identified as to type, caliber and model, as well as with a lot number and the supplier's identification as assigned by the procuring activity. Each lot shall be further identified by a Federal Stock Number assigned by the procuring activity.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. The AQL for the Major Class shall be 0.25 percent and the AQL for the Minor Class shall be 1.50 percent.

4.3.2.1 Classification of defects.- The classification of defects shall be as specified in Table I.

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TABLE I

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Visual 1/				
	Cartridge				
1	Discolored, dirty, oily, smeared			X	
2	Corroded; or stained, if etched		X		
	Case				
4	Round head		X		
5	Dent				X
6	Split case				
	in K, L or M location	X			
	in I, S or J location		X		
7	Perforated case	X			
8	Draw scratch				X
9	Scratch			X	
10	Beveled underside of head		X		
11	Case mouth not crimped in cannelure		X		
12	Scaly metal				X
13	No chamfer on head (rim)		X		
14	Fold			X	
15	Wrinkle			X	
16	Buckle			X	
17	Bulge			X	
18	Illegible or missing head stamp			X	
19	Defective head			X	
20	Defective mouth			X	
21	No visible evidence of mouth anneal		X		
	Bullet				
22	Dent			X	
23	Scratch			X	
24	Split bullet jacket		X		
25	Loose bullet		X		
26	Missing cannelure		X		
27	Scaly metal				X
28	Upset (crooked) point			X	
31	Defective cannelure			X	
	Primer				
32	No primer	X			
33	Cocked primer	X			
34	Inverted primer	X			
35	Loose primer		X		
36	Nicked or dented primer			X	

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TABLE I (Cont'd)

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Primer, cont.				
37	No waterproofing material (primer pocket joint)			X	
38	Defective crimp			X	
	Gaging				
39	Total length		X		
40	Cartridge profile failure (requiring more than 80 lbs. dead weight to insert in profile and alignment gage)		X		
41	Diameter of extractor groove, max.		X		
42	Diameter of extractor groove, min.			X	
43	Diameter of head		X		
44	Thickness of head		X		
45	Length to shoulder datum		X		
46	Depth of primer		X		
	Weighing				
47	Weight, min. 2/		X		

1/ Refer to MIL-STD-636 for visual defect standards for defects 1 through 38.

2/ Each lightweight cartridge shall be disassembled and the propellant weighed. Each such cartridge found to contain less than 100 grains of propellant shall be classed as a critical defect.

4.3.2.2 Flash tube.- The debulleted cases obtained from the bullet extraction test shall be examined for "missing flash tube" and "lacquer seal on flash tube missing or punctured." The occurrence of either defect in any cartridge case shall be cause for rejection of the lot.

4.3.3 Tests.- The tests listed in Table II shall be conducted in accordance with the methods and procedures specified in 4.4.

4.3.3.1 Test samples.- The quantities for the various tests shall be as specified in Table II. Only cartridges having met the visual, dimensional and weight requirements shall be used in the ballistic tests and shall have been selected in such a manner that the sample is representative of the entire lot. The cartridges shall be thoroughly mixed before being divided into samples for the various tests.

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TABLE II

Test	Number of Cartridges	Requirement Paragraph
Bullet extraction 1/	25	3.3
Residual stress (Mercurous nitrate) 1/	50	3.4
Waterproof 2/	20	3.5
Velocity 3/	20	3.6
Chamber pressure 3/	20	3.7
Accuracy 3/	90	3.8
Function & casualty 4/		
Rifle, Spotter, Cal. .50, M8C	200	3.9

1/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test, a second sample consisting of double the number of cartridges in the first sample may be tested. If any failing cartridges are found in the second sample, the lot shall be rejected.

2/ Failure of seven or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If more than three but less than seven cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. The lot shall be rejected if in the combined first and second sample seven or more cartridges fail to comply with the applicable requirement.

3/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges used in the first test.

Failure of the cartridges in the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

4/ The lot shall be rejected when function and casualty defects plus all other firing defects exceed the acceptance number for the cumulative sample in Table III. If the number of defects found in the first tests exceeds the acceptance number for the first sample, but is equal to or less than the acceptance number for the cumulative sample, a second sample, consisting of double the quantities specified under function and casualty test, shall be fired. If the total number of defects in the combined first and second sample exceeds the acceptance number for the cumulative sample the lot shall be rejected. If, in testing a second sample, defects other than those for which the second sample is being tested should occur to the extent that they exceed the acceptance number for the cumulative sample, the lot shall be rejected.

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4.3.3.2 Firing defects.- Firing defects and acceptance numbers shall be as specified in Table III.

TABLE III

Defects	Acceptance Number First Sample	Acceptance Number Cumulative (1st & 2nd Sample)
1. Misfire	1	2
2. Bullet remaining in bore 1/	0	-
3. Primer leak		
a. Perforation in firing pin indent in primer cup	1	2
b. Escape of gas through primer cup other than 3a	1	2
c. Escape of gas around primer cup more than 50% of periphery	3	5
d. Blown primer or primer falls out of pocket on retraction of bolt	0	1
e. Primer remains in pocket but is physically loose	2	3
4. Case casualties		
a. Longitudinal split 2/		
(1) Neck and shoulder (I or S)	8	14
(2) Body (J)	2	3
(3) Body (K)	1	2
(4) To head (L)	0	1
(5) Through head (M)	0	1
b. Circumferential rupture 2/		
(1) Partial, shoulder or body (J&S)	2	3
(2) Partial, body (K)	0	1
(3) Partial, head (L)	0	1
(4) Complete	0	1
c. Flash tube		
(1) Missing flash tube 3/	0	1
(2) Flash tube remaining in bore 1/	0	-
5. Failure to extract	0	1
6. Weapon stoppage 4/	0	1

1/ No second sample permitted. Lot shall be rejected.

2/ For location of defects indicated by letters in parentheses, see Drawing C7643674.

3/ A loose flash tube shall carry no penalty.

4/ All stoppages attributable to the ammunition, with the exception of misfire, complete rupture or failure to extract, observed in all tests shall be included.

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4.3.4 Packaging, packing, and marking inspection.- During or immediately prior to the packaging operation, 100% examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of a high pressure test cartridge shall be classed as a critical defect. Occurrence of any incorrect type other than high pressure test shall be classed as a major defect. All nonconforming cartridges shall be rejected. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644, as applicable to the drawing.

4.3.5 Inspection equipment.- The examination and tests shall be made using the equipment listed on IEL-10534667.

4.4 Test methods and procedures.

4.4.1 Bullet extraction.- The cartridges shall be tested in an approved bullet extractor machine. The rate of travel of the test head shall be not less than three nor more than six inches per minute.

4.4.2 Residual stress (mercurous nitrate).- The test shall be conducted in accordance with AMCR 700-700, Vol. III.

4.4.3 Waterproof.- The ammunition to be tested, not to exceed five cartridges at one time, shall be placed horizontally on a perforated tray. The tray shall be placed in a desiccator containing a sufficient quantity of freshly boiled water to maintain a head of 2 to 2½ inches. The desiccator shall be evacuated to a pressure of 7½ pounds per square inch (15 inches of mercury) below atmospheric pressure and held at that pressure for 30 seconds. The number of bubbles liberated from the mouth or primer of each cartridge shall be observed.

4.4.4 Velocity.- The test shall be conducted in accordance with AMCR 700-700, Vol. III.

4.4.5 Chamber pressure.- The test shall be conducted in accordance with AMCR 700-700, Vol. III.

4.4.6 Accuracy.- The test shall be conducted in accordance with AMCR 700-700, Vol. III.

4.4.7 Function and casualty.- The test shall be conducted in accordance with AMCR 700-700, Vol. III. Two rifles shall be used and 100 cartridges shall be fired in each.

4.4.8 Defect penalty.- In any ballistic test, except function and casualty, in which the occurrence of a firing defect prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. That particular test shall not be penalized, but the acceptance or initial production sample shall be penalized for such defects in accordance with Table III.

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5. PREPARATION FOR DELIVERY

5.1 Packing.- Level A (Worldwide shipment) - The cartridges shall be packed in accordance with Drawing F10534780.

5.2 Marking and labeling.- Packing boxes shall be marked and labeled in accordance with the drawing cited in 5.1.

6. NOTES

6.1 Ordering data.- Invitations for bids or request for proposal and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Type and level of packing.

6.1.3 Provisions for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.1.4 Provisions for the submission of acceptance of inspection reports containing final inspection results for each lot of ammunition presented to the Government.

Custodian:

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Preparing Activity:

Army - MU

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